

# FATIMA JINNAH WOMEN UNIVERSITY



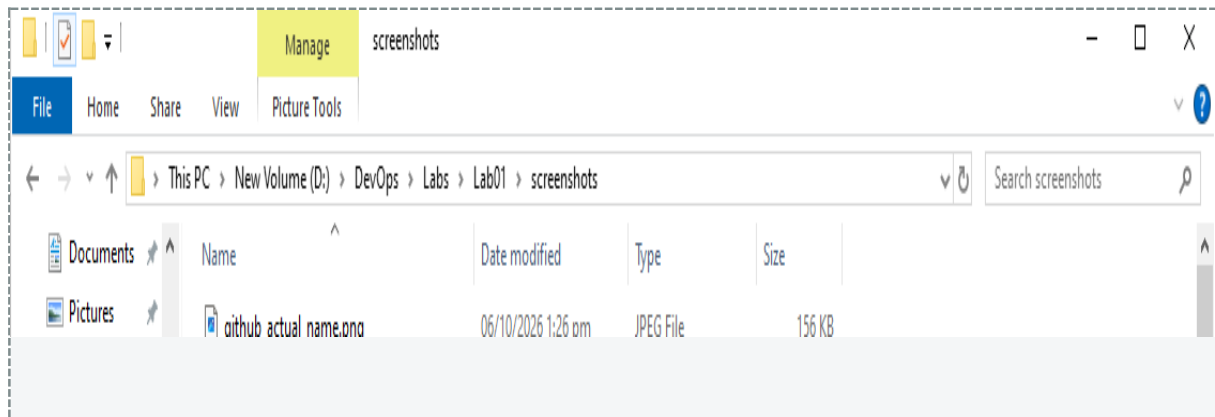
## **DevOps** **Lab 01**

GitHub Identity, Student Portal Registration, and Ubuntu Server  
Installation

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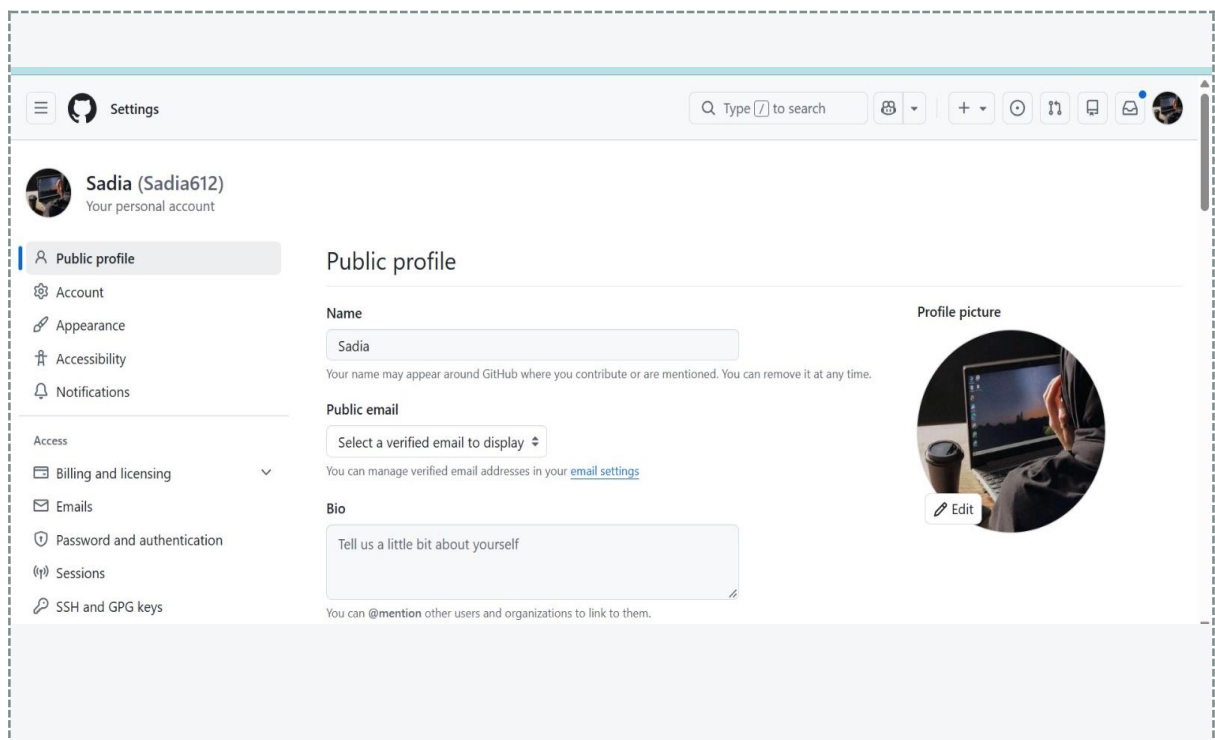
|                            |                 |
|----------------------------|-----------------|
| <b>Full Name</b>           | Sadia           |
| <b>Registration Number</b> | 2023-BCS-075    |
| <b>Course Code</b>         | DevOps_F26      |
| <b>Section (A / B)</b>     | A               |
| <b>GitHub Username</b>     | Sadia612        |
| <b>Instructor</b>          | Mr Waqas Saleem |

## Getting Started: Folder Structure

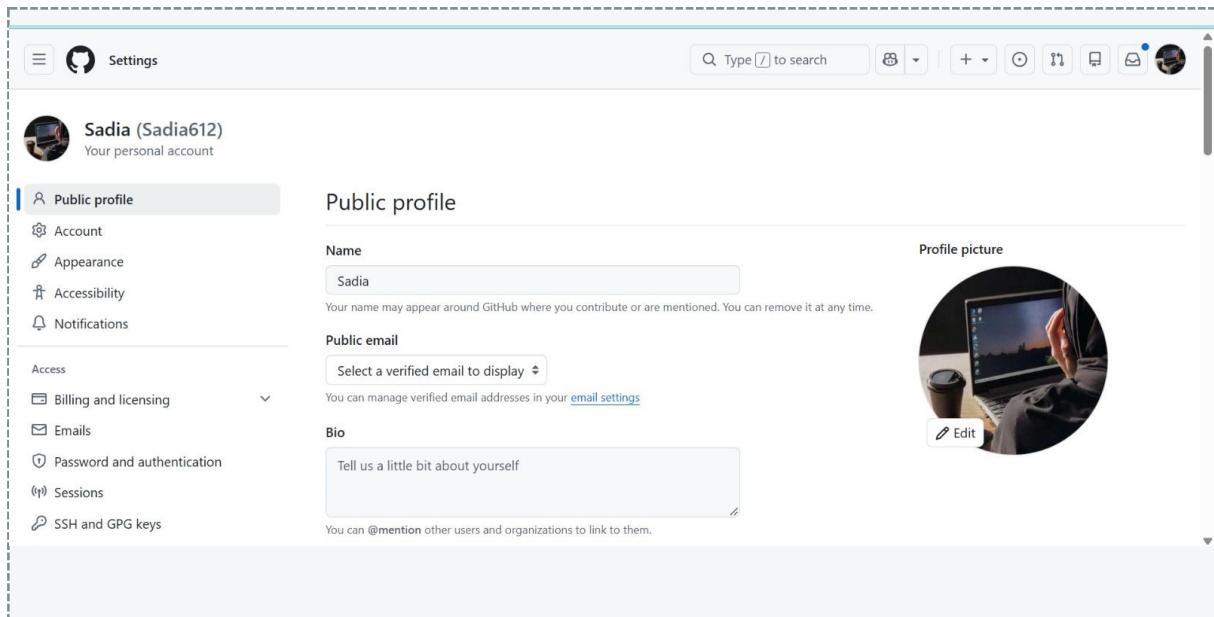


*Figure: Directory structure Labs/Lab01/screenshots/ created inside the DevOps repository.*

## Task 1: Create and Configure Your GitHub Account



*Figure: GitHub profile showing my profile name and GitHub username.*

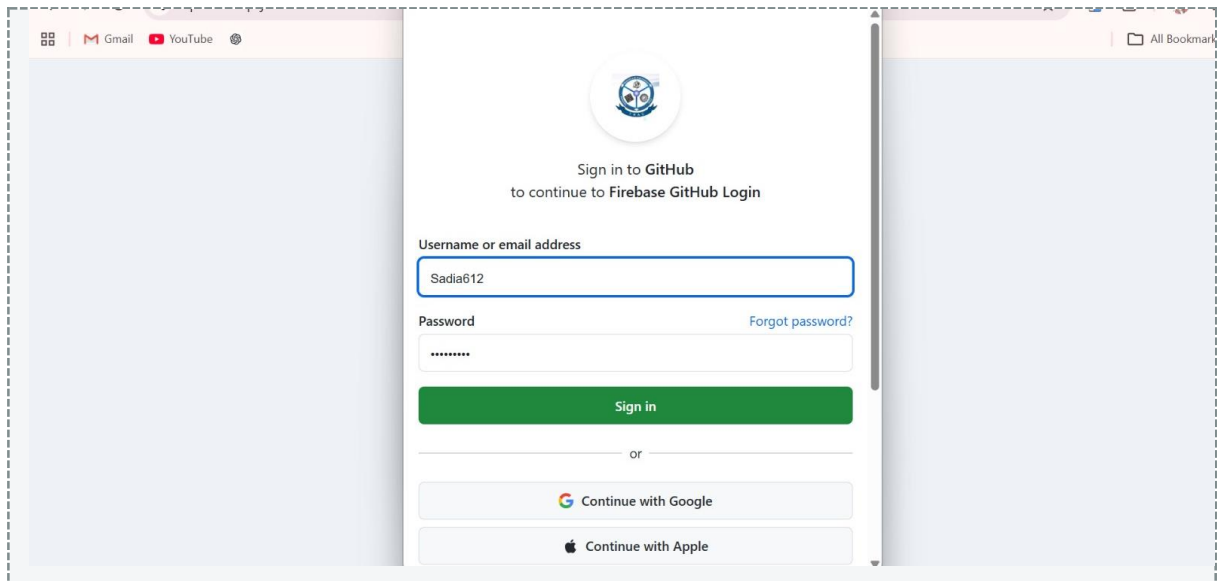


*Figure: GitHub Settings > Public profile with my complete actual name in the Name field.*

## Task 2: Register in the Student Marks Portal

The screenshot shows the Student Marks Portal login page. The page has a light blue background. In the center, there is a registration form. At the top of the form, it says 'Enter your registration number, course and section. The enrollment requires approval.' Below this, there are three dropdown menus: 'Registration number' (filled with '2023-BSCS-75'), 'Course' (filled with 'CS-649 — Cloud DevOps'), and 'Section' (filled with 'Section A'). Below the dropdowns, there is a dark blue button labeled 'Submit enrollment' and a light blue button labeled 'Sign out'. At the bottom of the form, there is a red text message: 'Enter a registration number in an enabled range shown above, including the required leading zeros.'

*Figure: Student Marks Portal login page.*

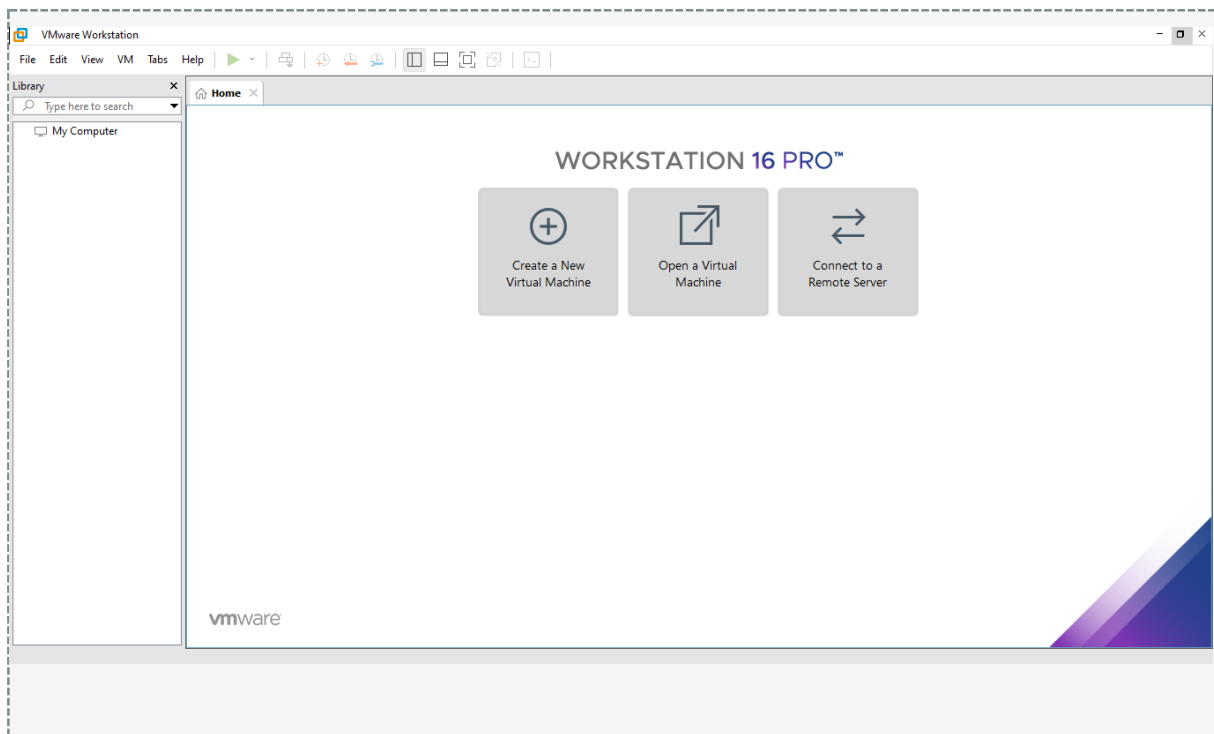


*Figure: My GitHub profile information shown inside the portal after signing in with GitHub.*

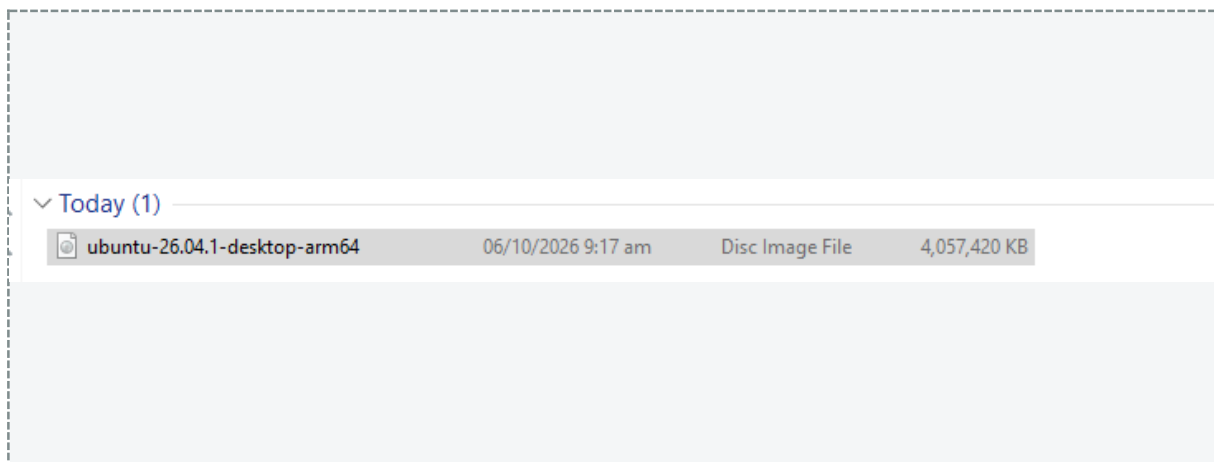


*Figure: Submitted/pending enrollment with my registration number, course code and section.*

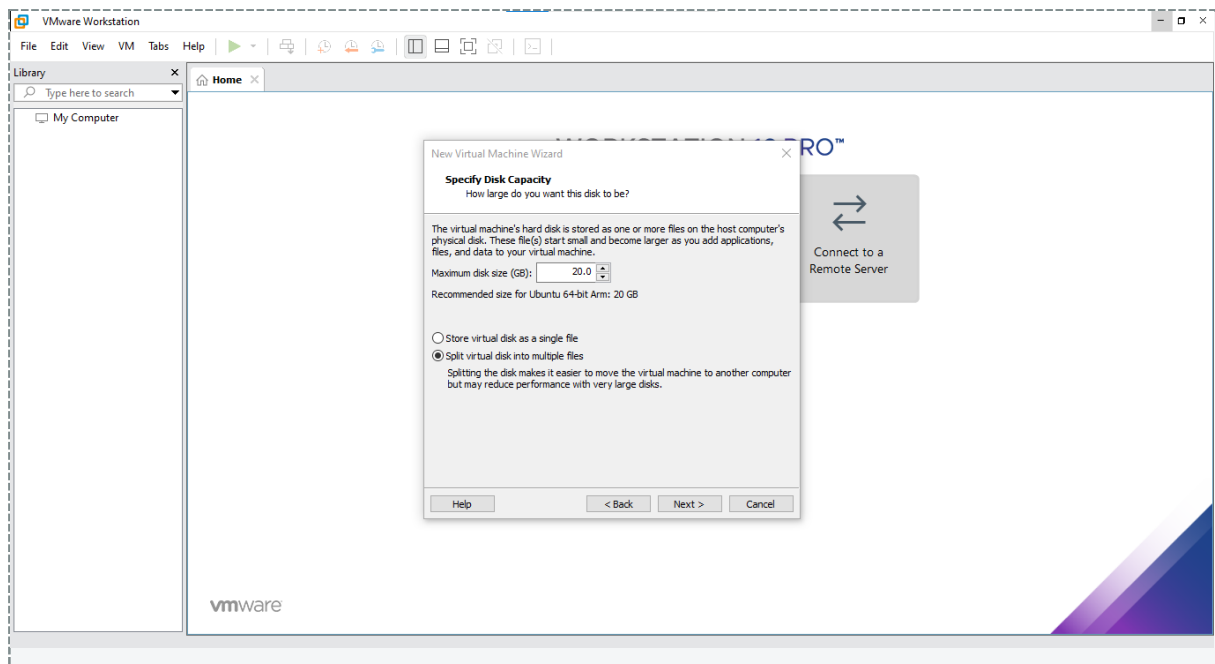
### Task 3: Download VMware Workstation Pro and Ubuntu Server



*Figure: VMware Workstation Pro installed and running.*

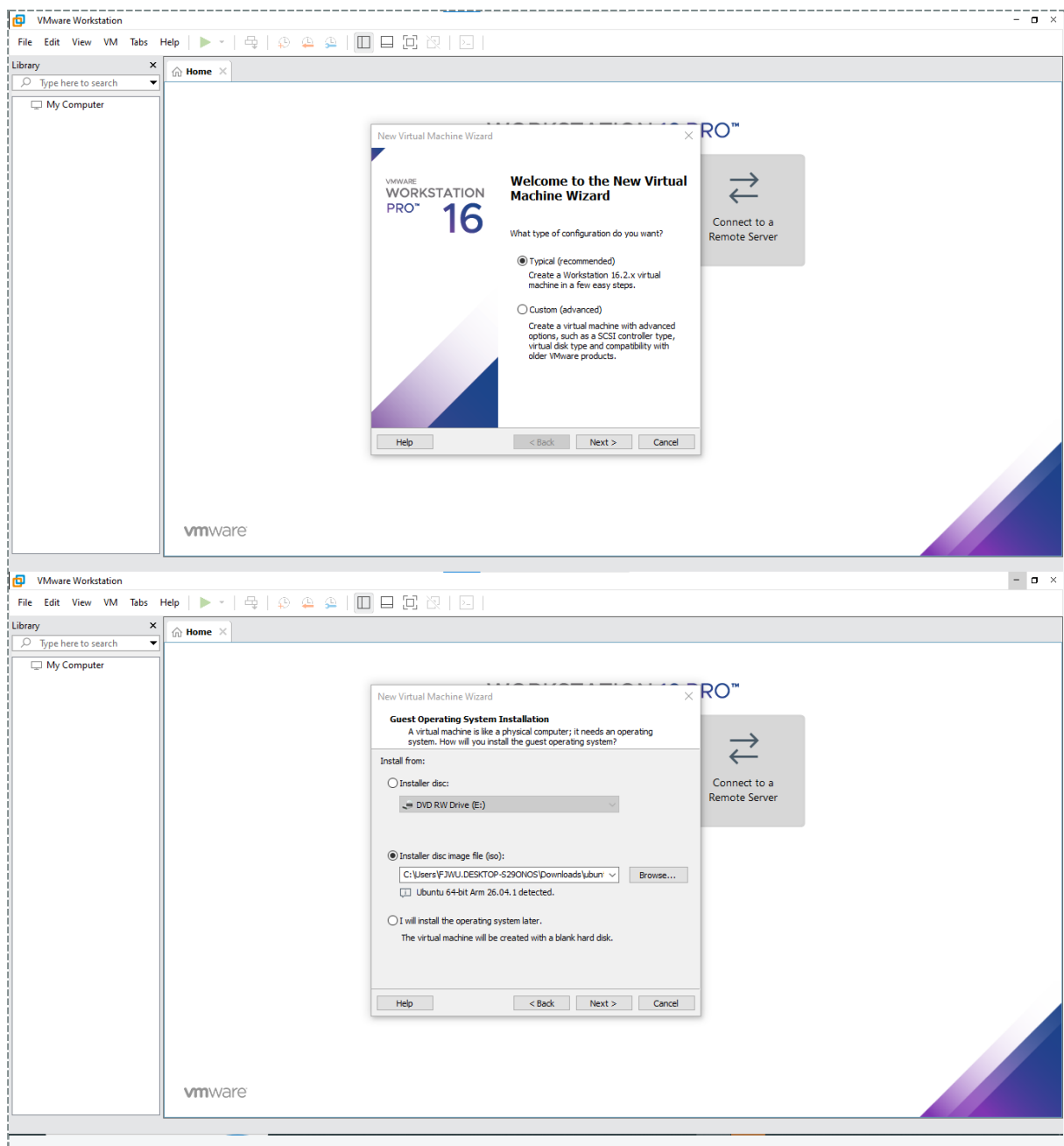


*Figure: Downloaded Ubuntu Server LTS ISO file with its filename visible.*

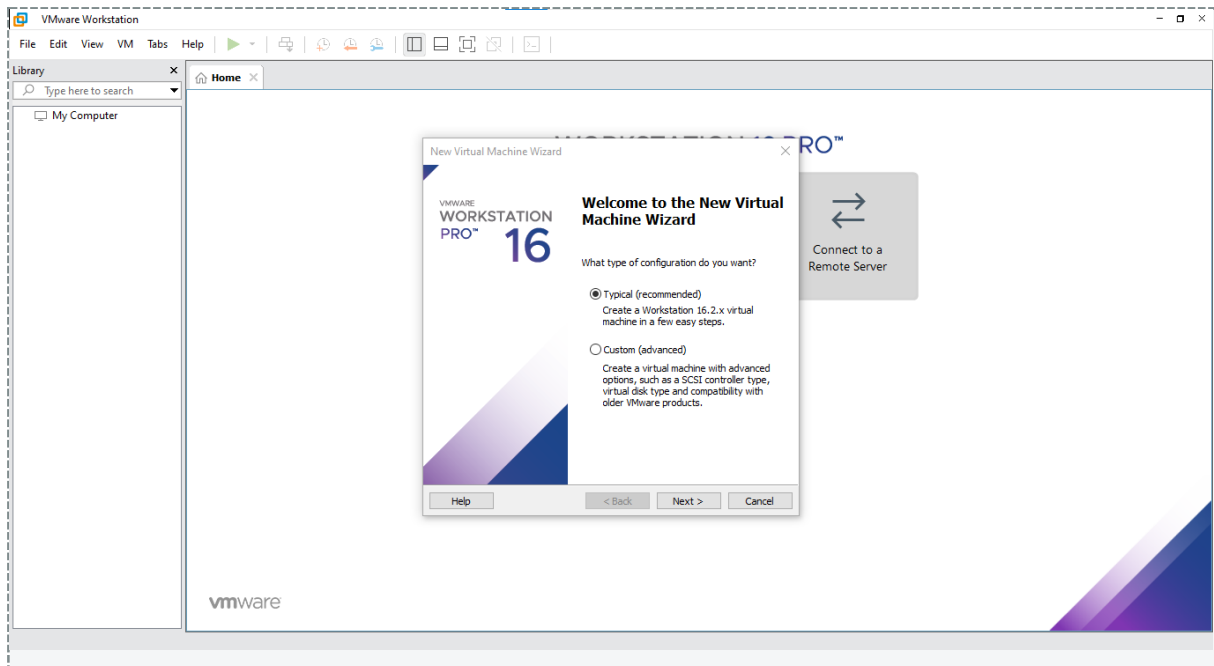


*Figure: Available free storage (at least 20 GB) for the virtual machine.*

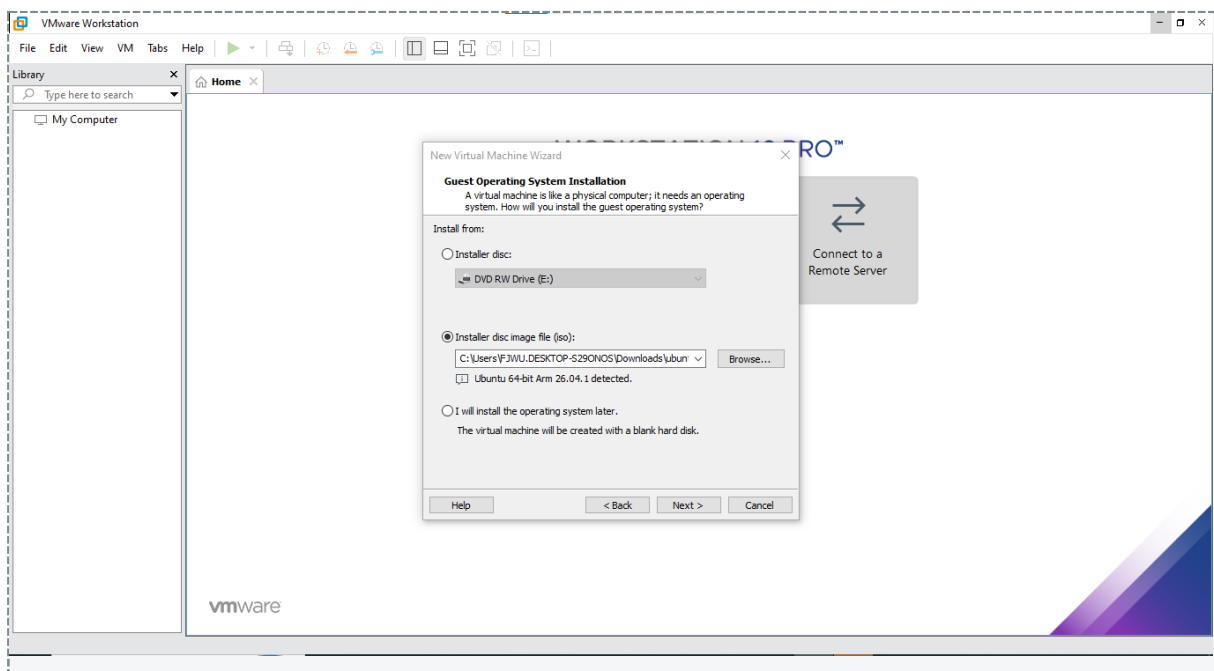
## Task 4: Create the Ubuntu Server Virtual Machine



*Figure: New Virtual Machine Wizard opened in VMware Workstation Pro.*

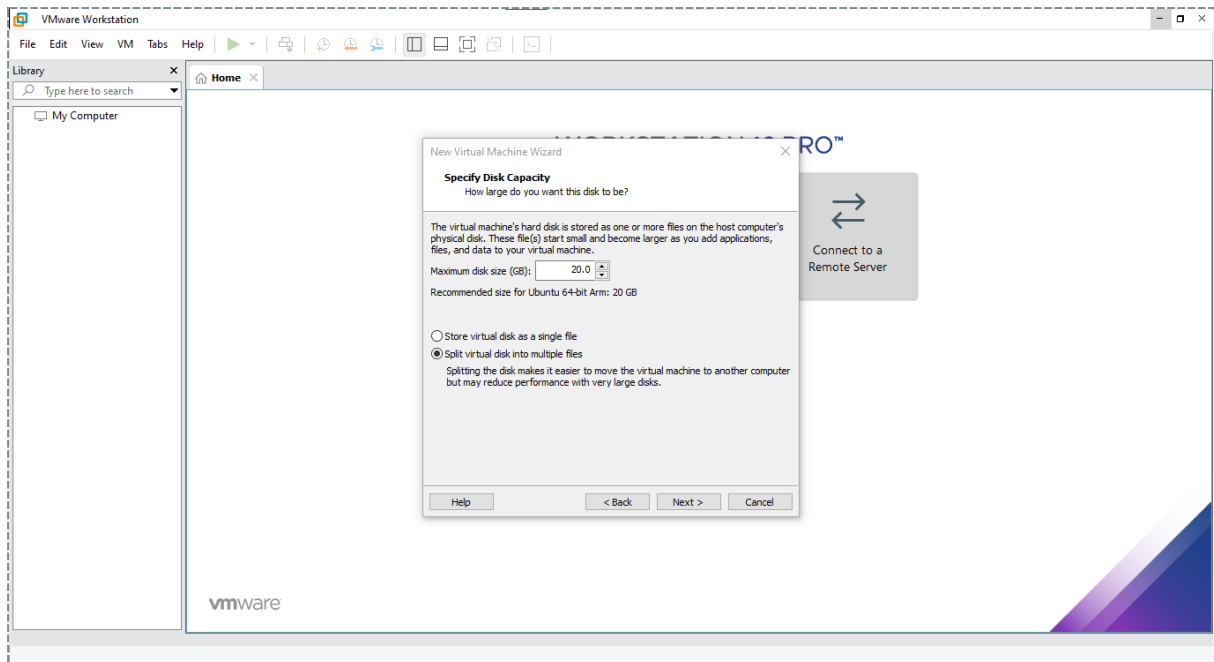


*Figure: Typical (recommended) configuration selected.*

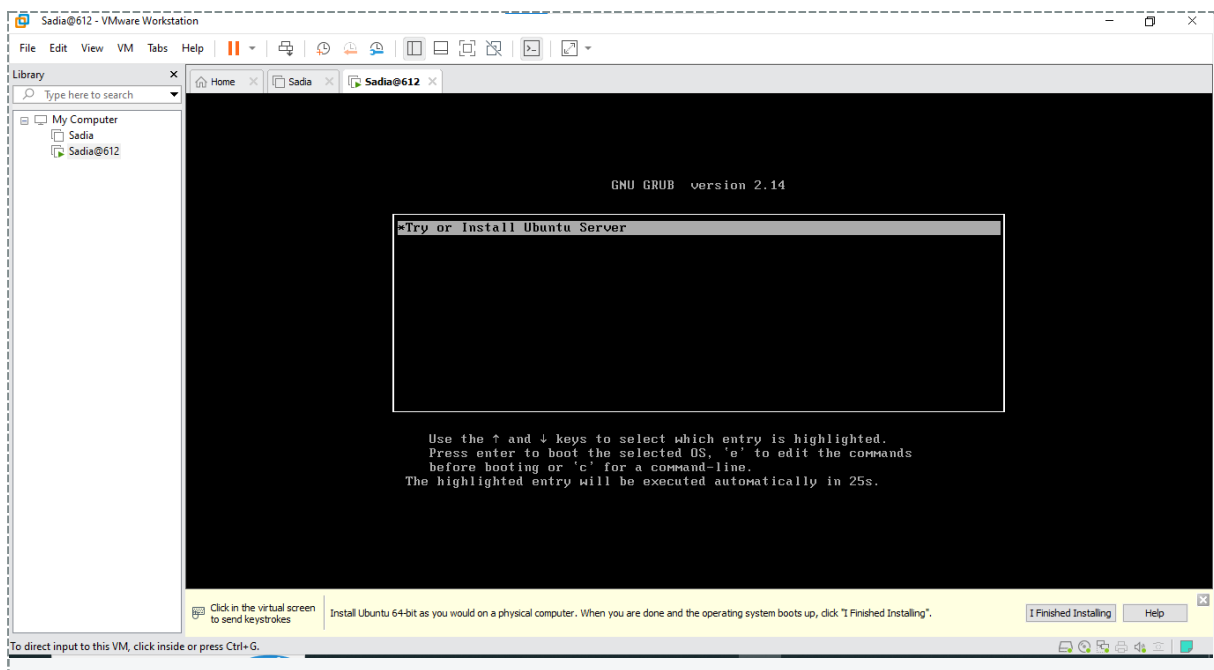


*Figure: Installer disc image file (ISO) selected with the Ubuntu Server ISO path.*



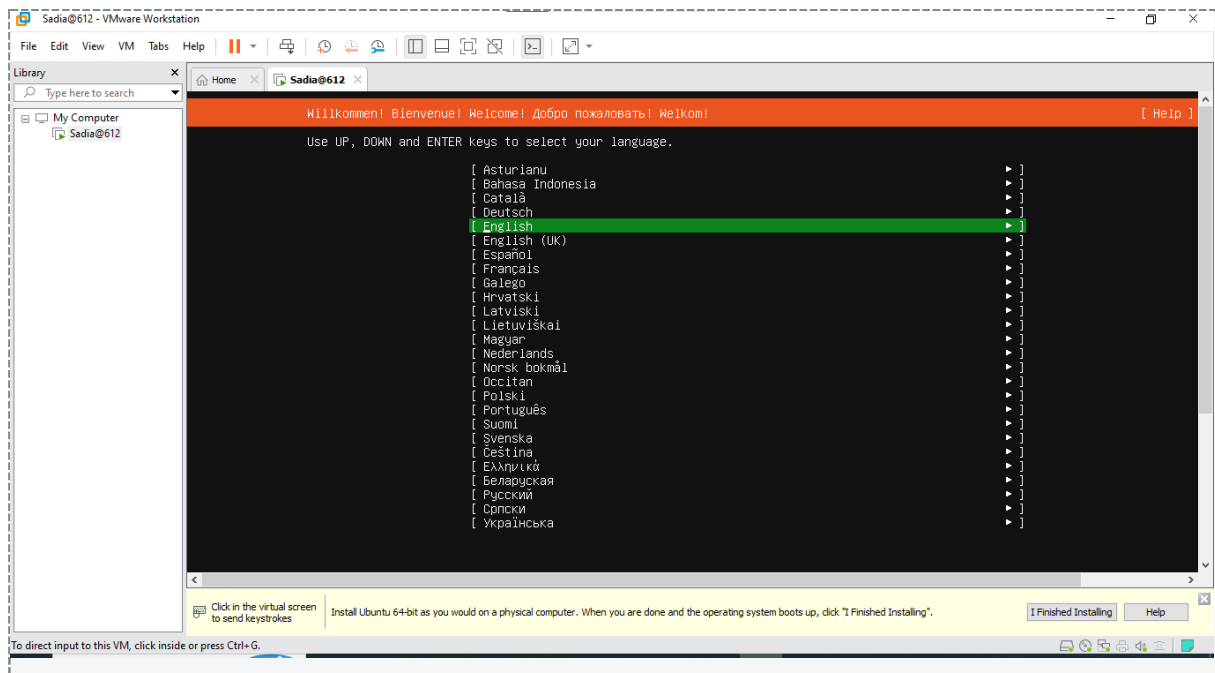


*Figure: Final virtual machine configuration summary with at least 20 GB disk.*



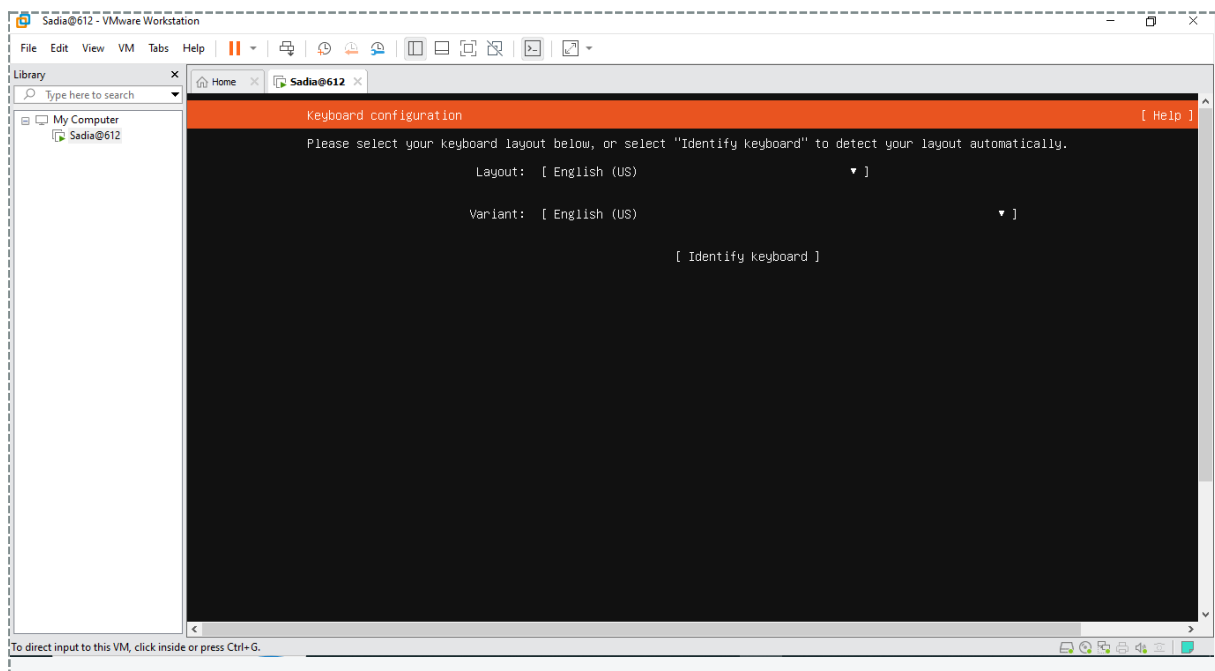
*Figure: Ubuntu Server installer boot screen after powering on the VM.*

## Task 5: Choose the Installation Language

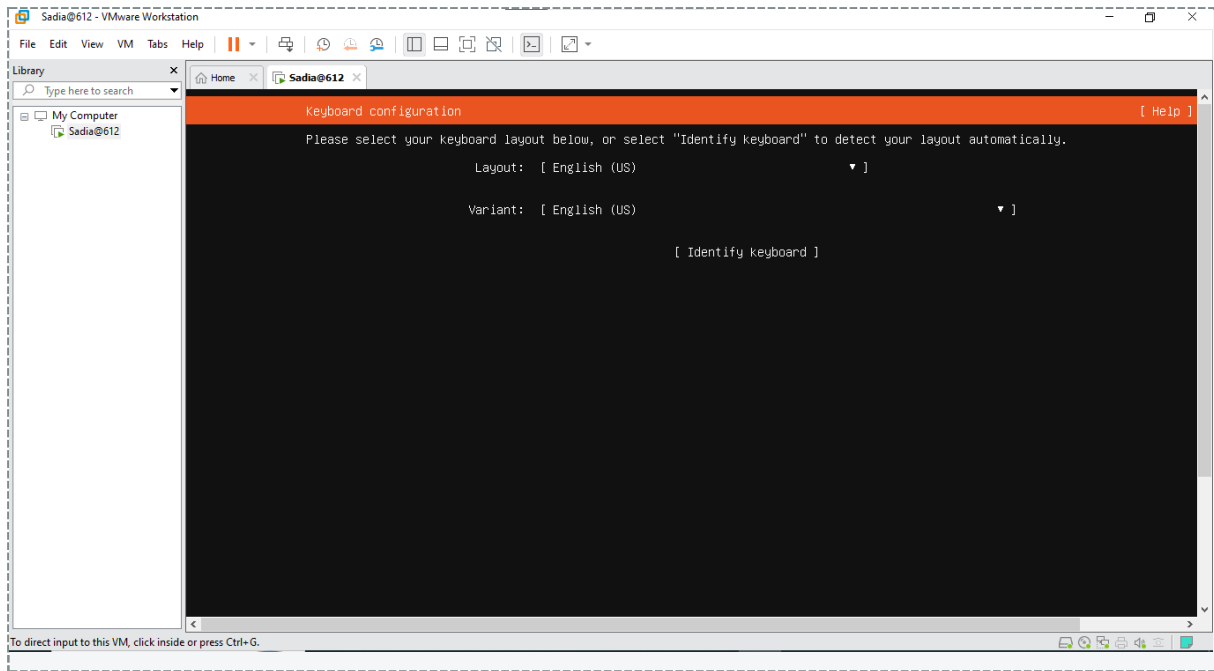


*Figure: Installation language selected in the Ubuntu Server installer.*

## Task 6: Configure the Keyboard Layout

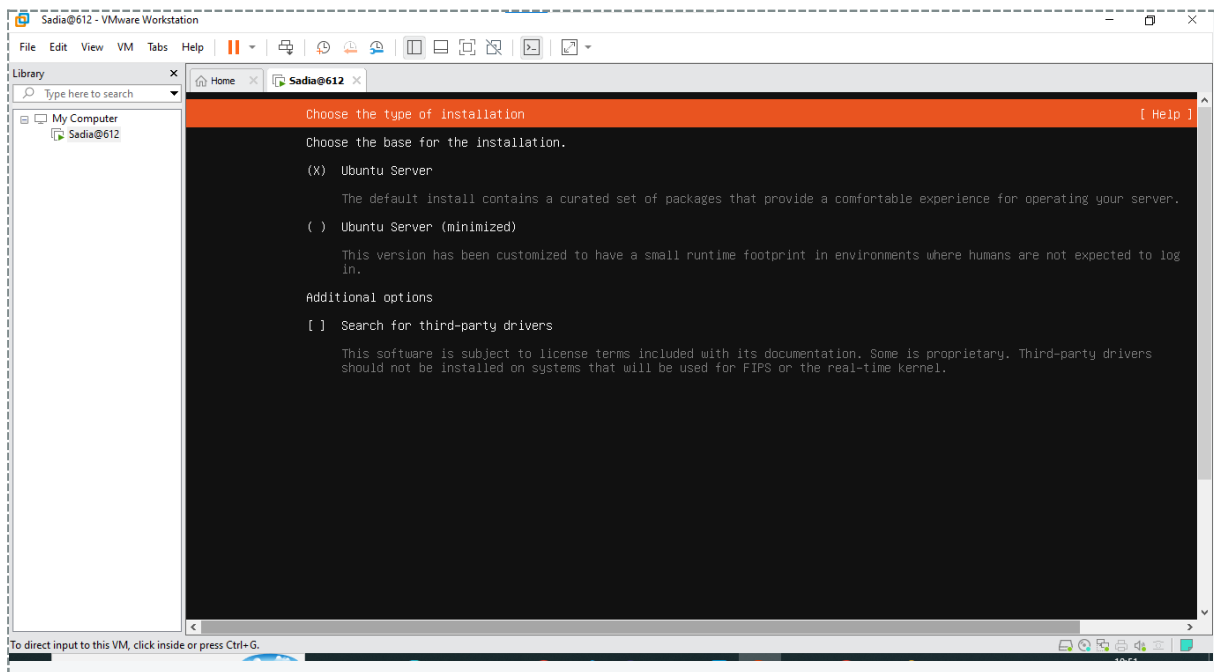


*Figure: Keyboard layout selected.*



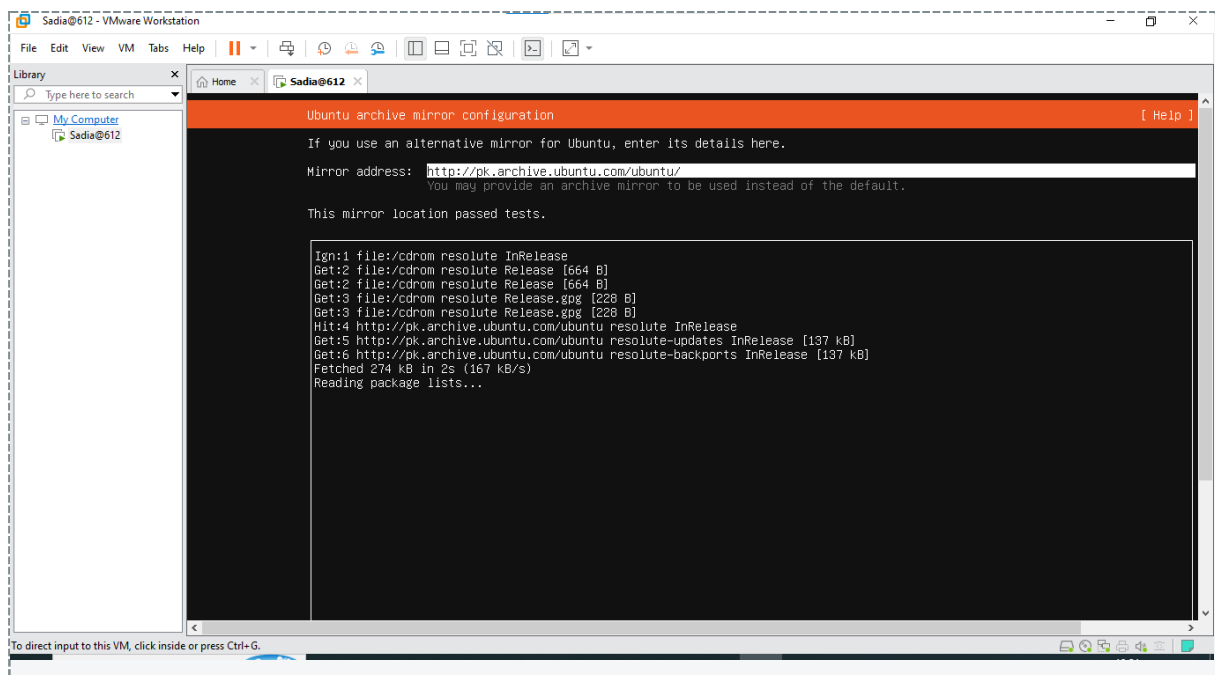
*Figure: Keyboard variant selected.*

## Task 7: Choose the Ubuntu Server Installation Type



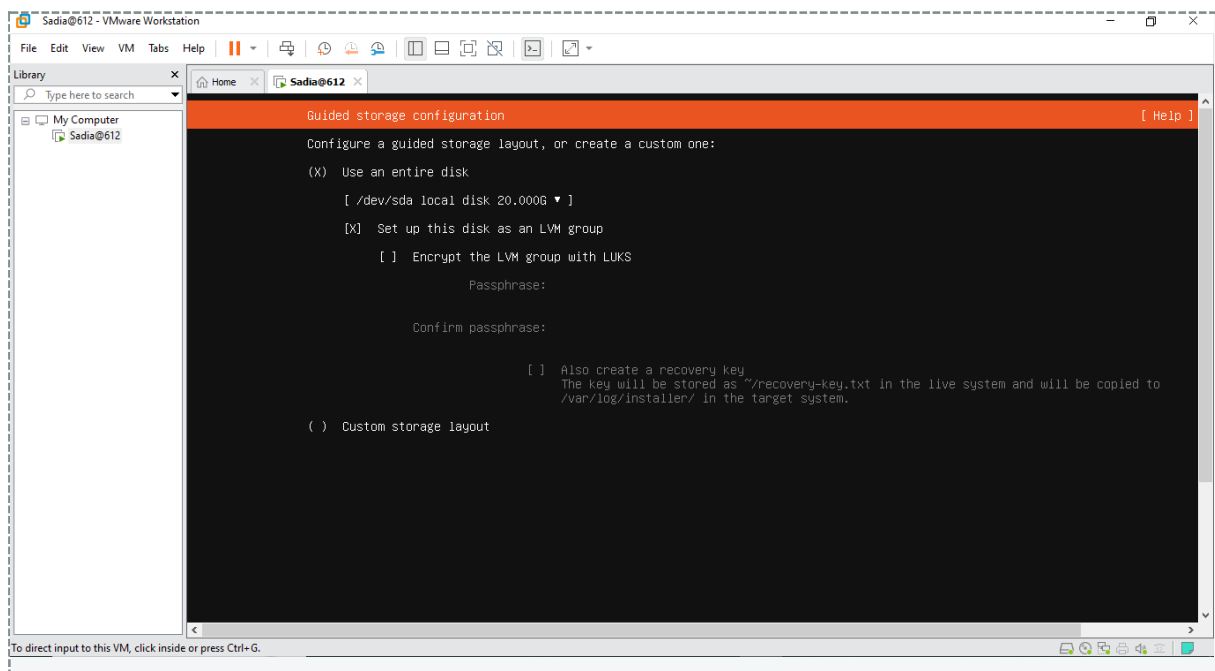
*Figure: Normal Ubuntu Server installation type selected (not MAAS).*

## Task 8: Configure the Network



*Figure: Network interface detected by the installer.*

## Task 9: Configure Storage



*Figure: Guided storage option selected to use the VMware virtual disk.*

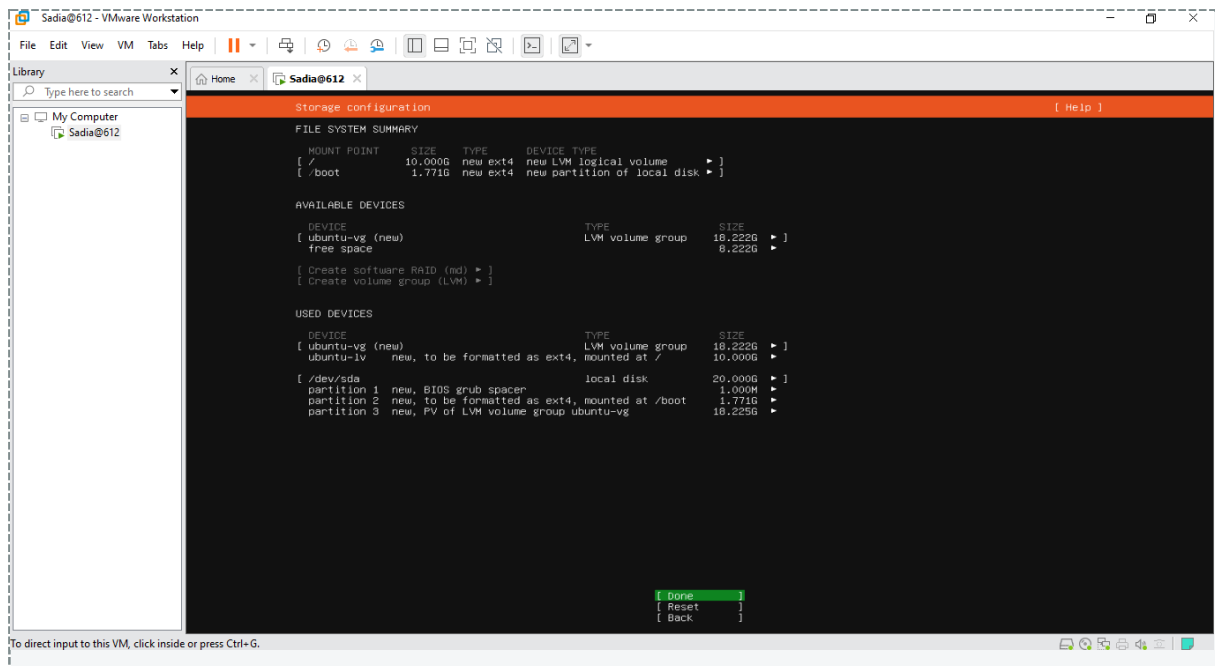


Figure: Storage configuration reviewed before continuing.

## Task 10: Select the Installation Disk

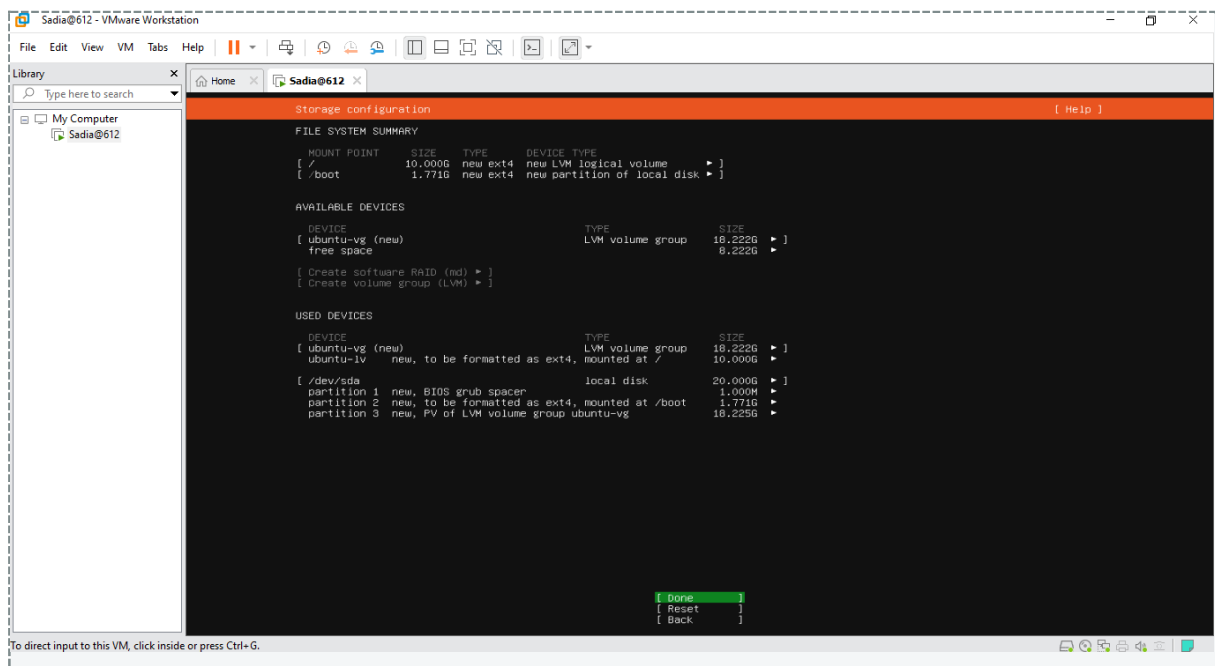
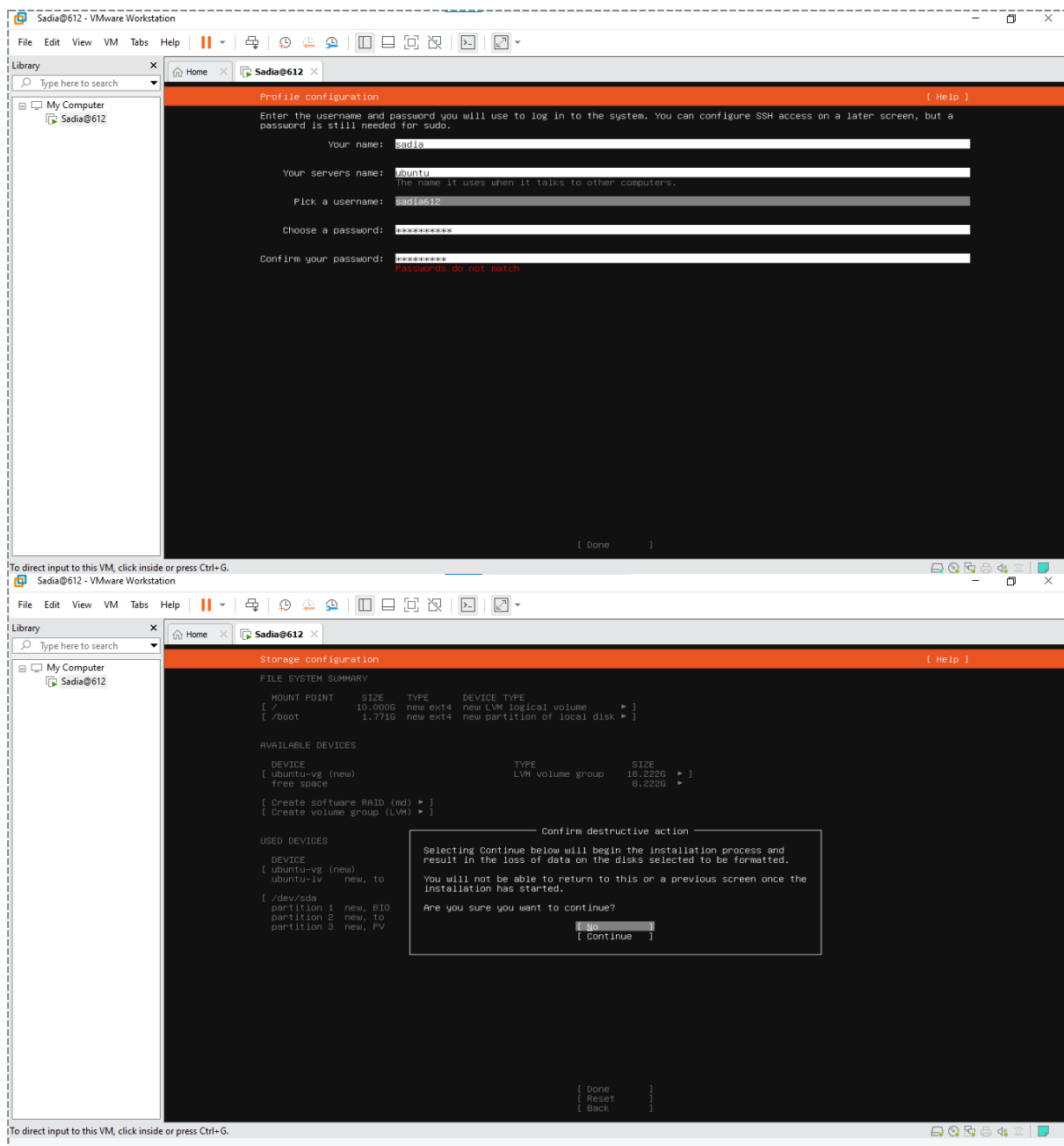
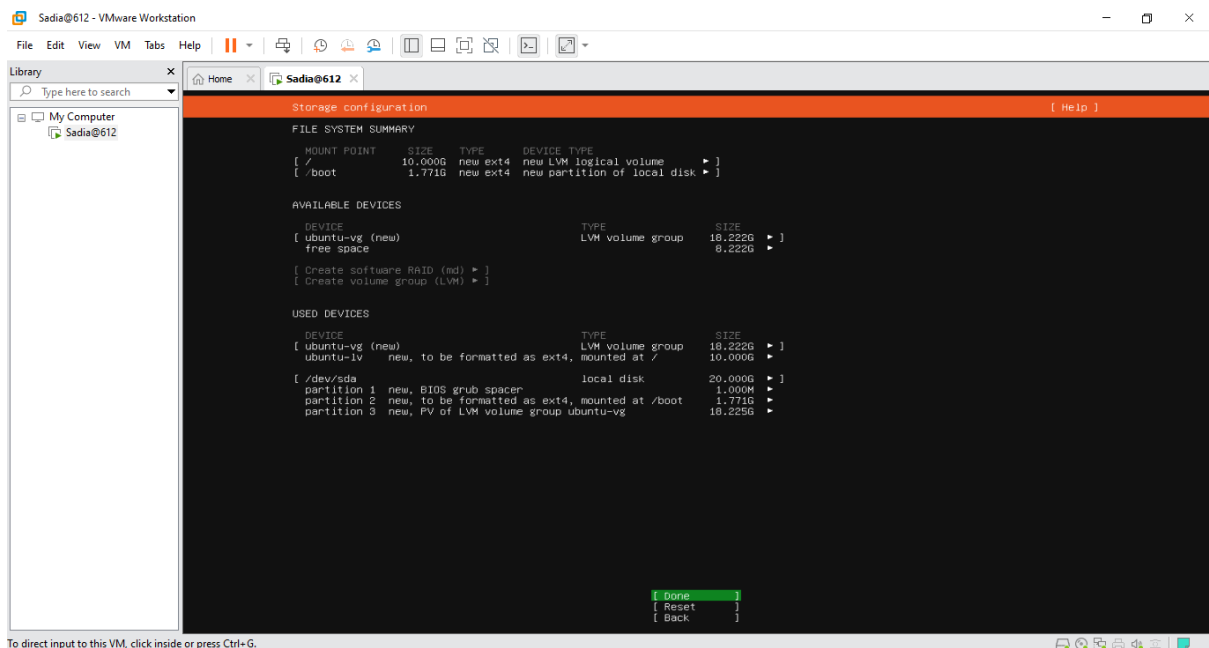


Figure: VMware virtual disk selected for the installation.

## Task 12: Confirm the Storage Changes

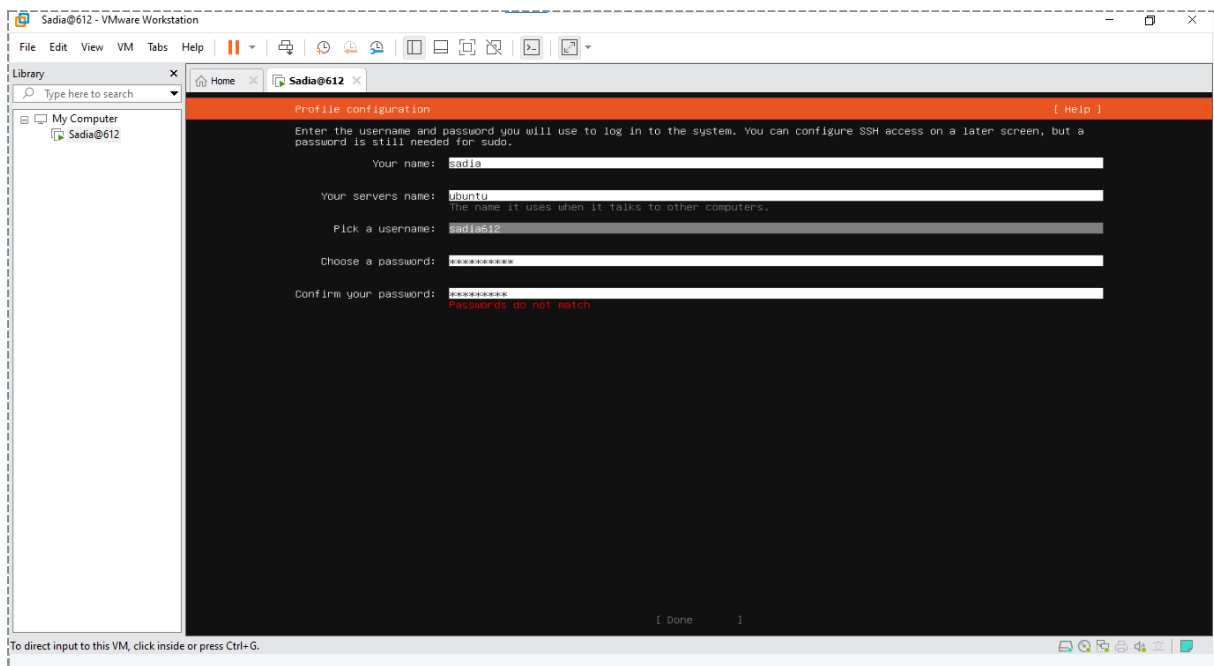


*Figure: Destructive-action warning shown before applying storage changes.*



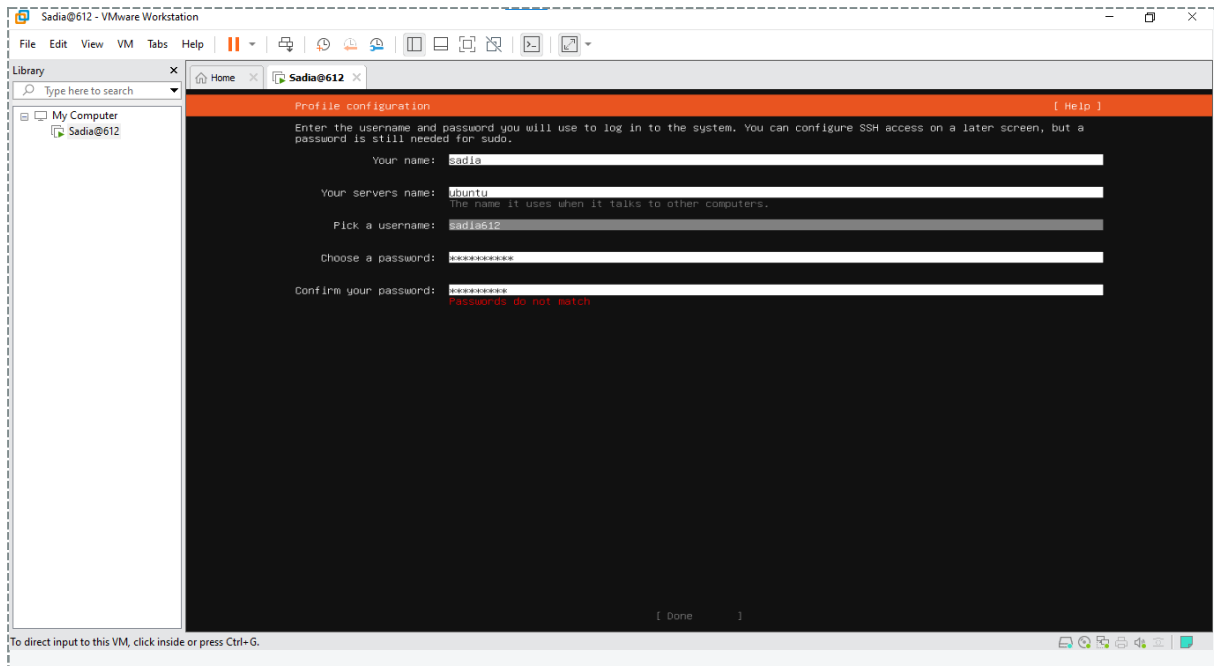
To direct input to this VM, click inside or press Ctrl+G.

## Task 13: Configure the Ubuntu Profile

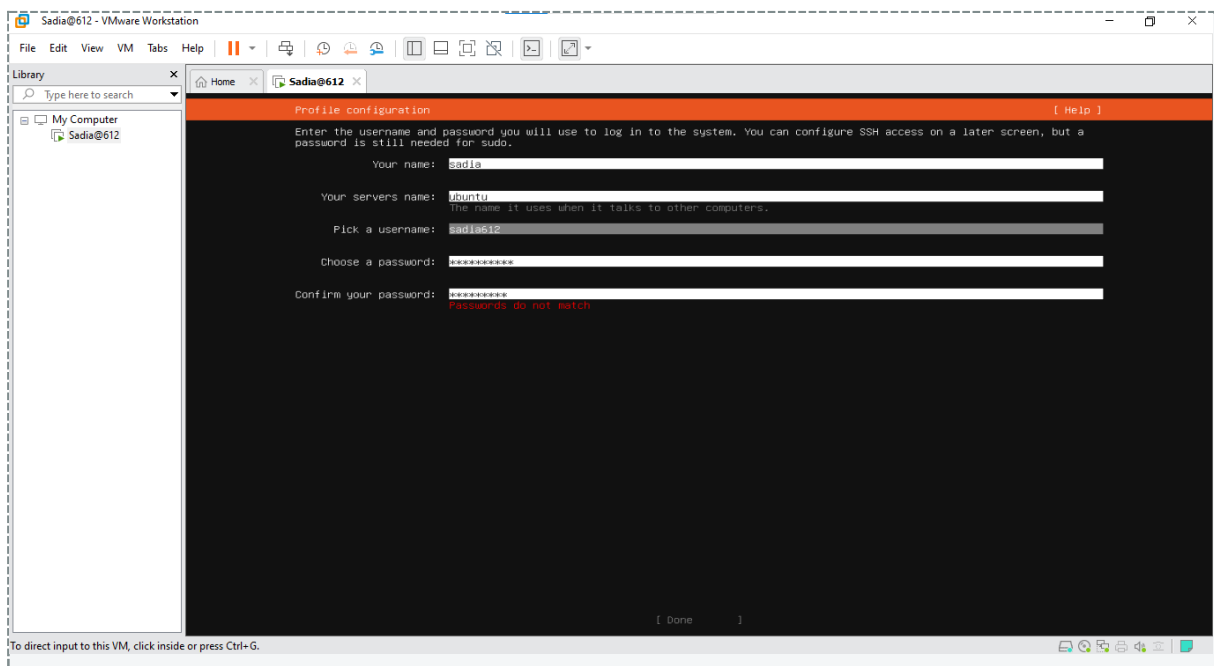


To direct input to this VM, click inside or press Ctrl+G.

*Figure: My complete actual name entered in the Your name field.*

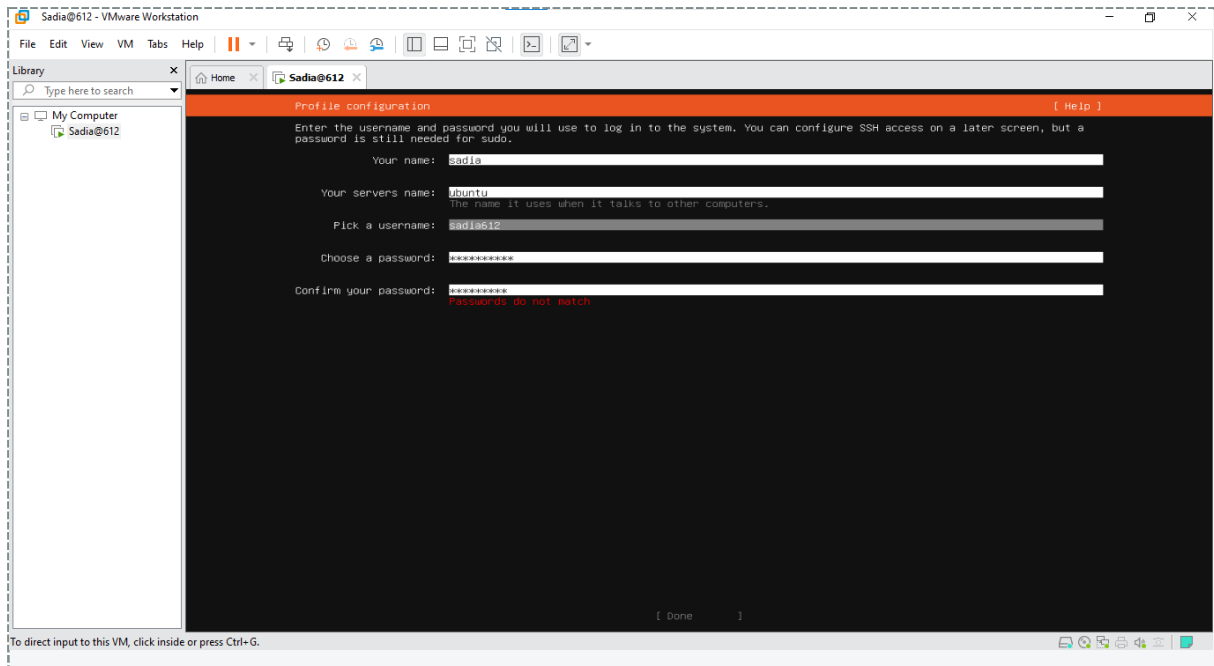


*Figure: Server name entered as ubuntu.*



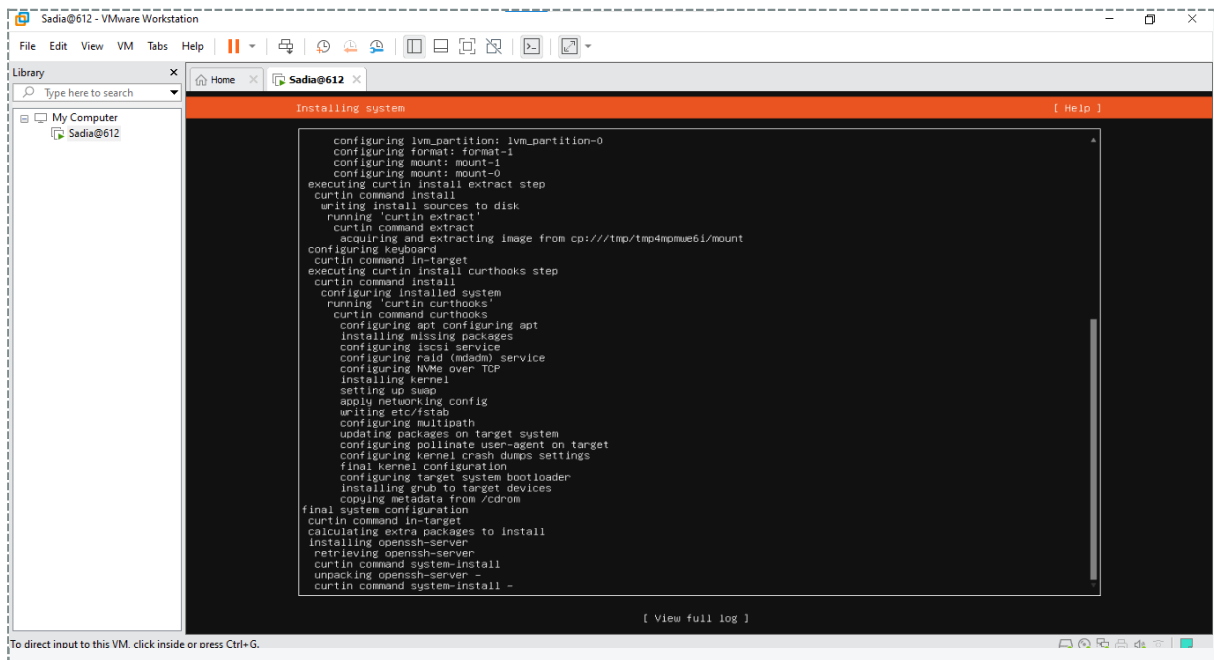
*Figure: Ubuntu username entered exactly as my GitHub username.*





*Figure: Completed profile page (password not exposed).*

## Task 14: Complete the Software Installation



*Figure: Ubuntu Server software installation in progress.*

## Task 15: Reboot and Verify the Ubuntu Identity

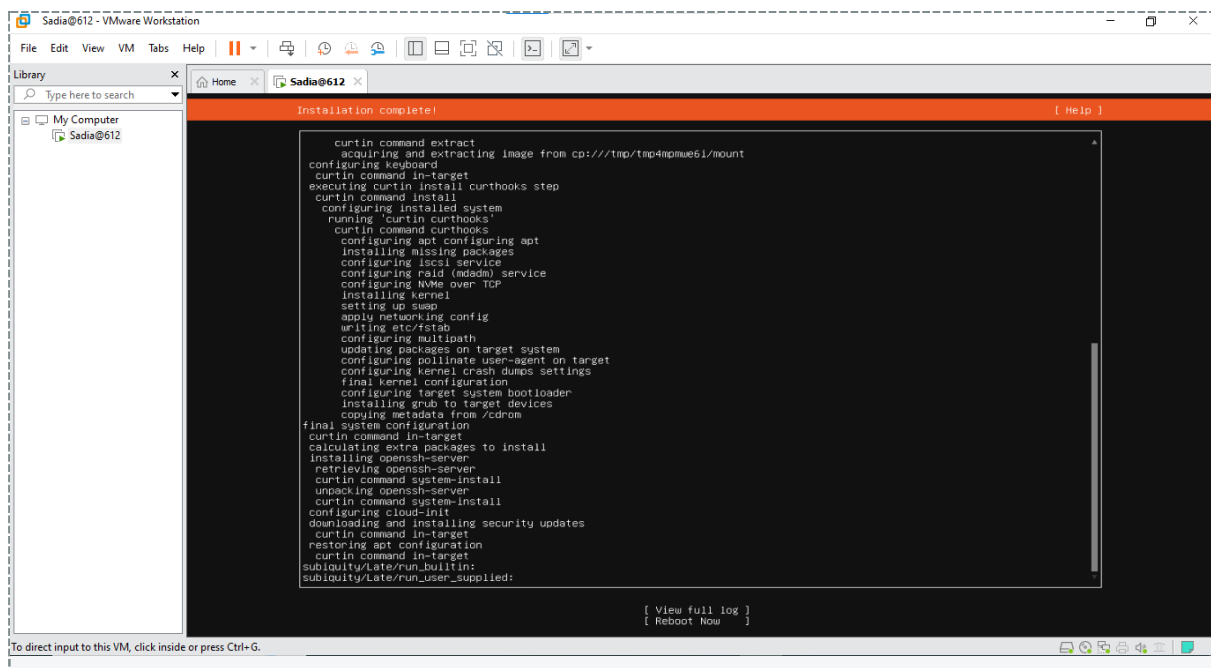


Figure: Installation complete screen with the Reboot Now option.

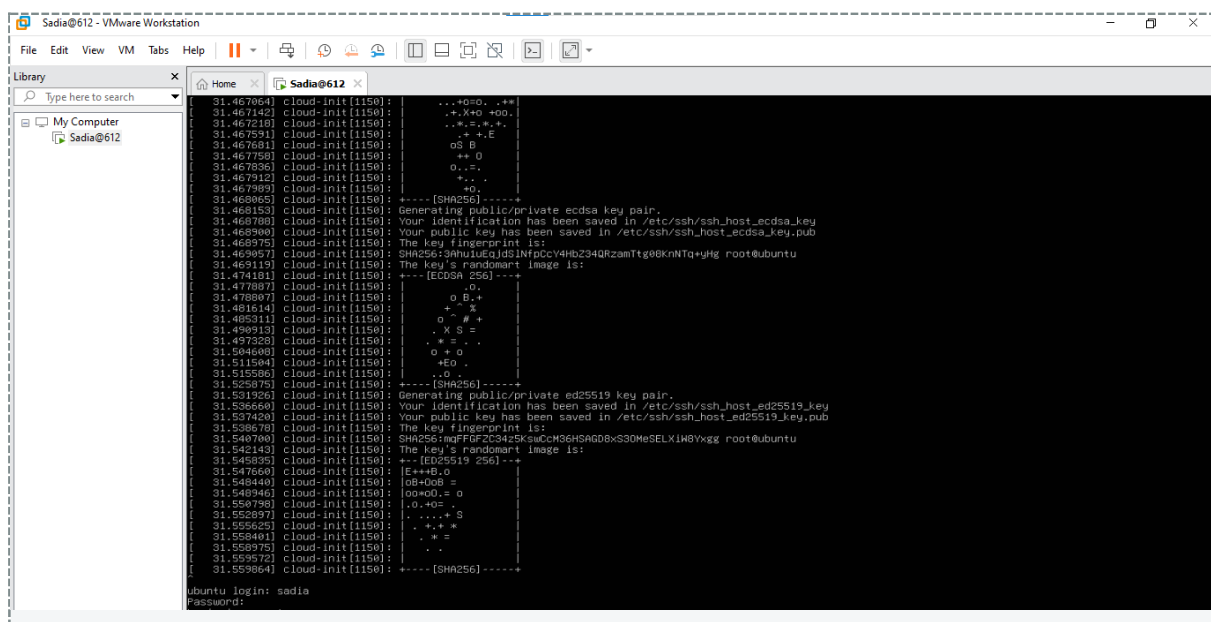
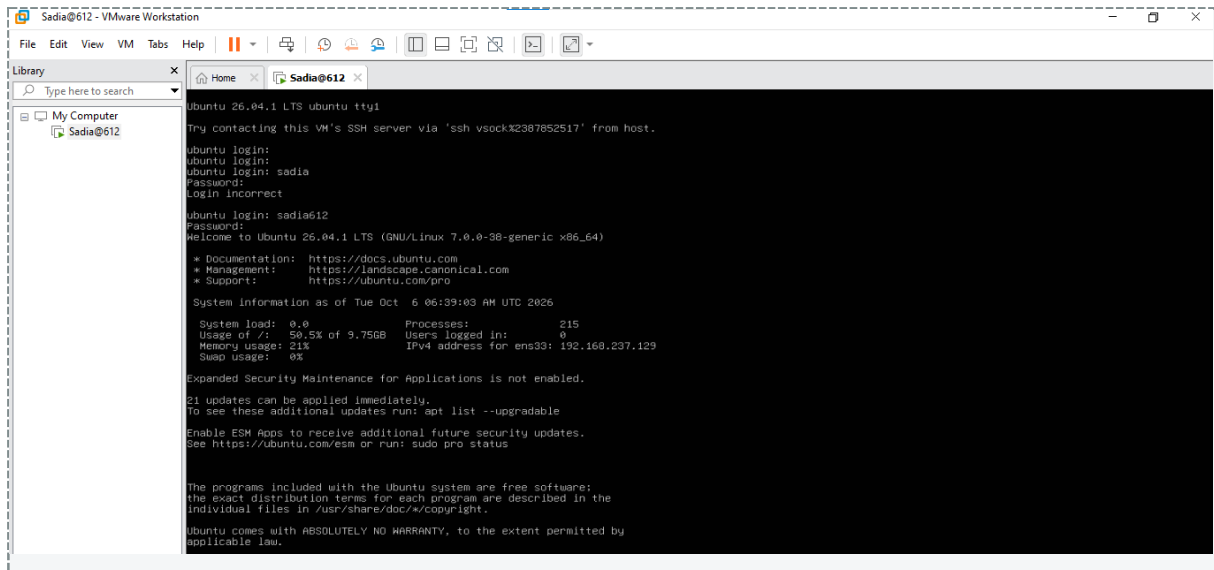
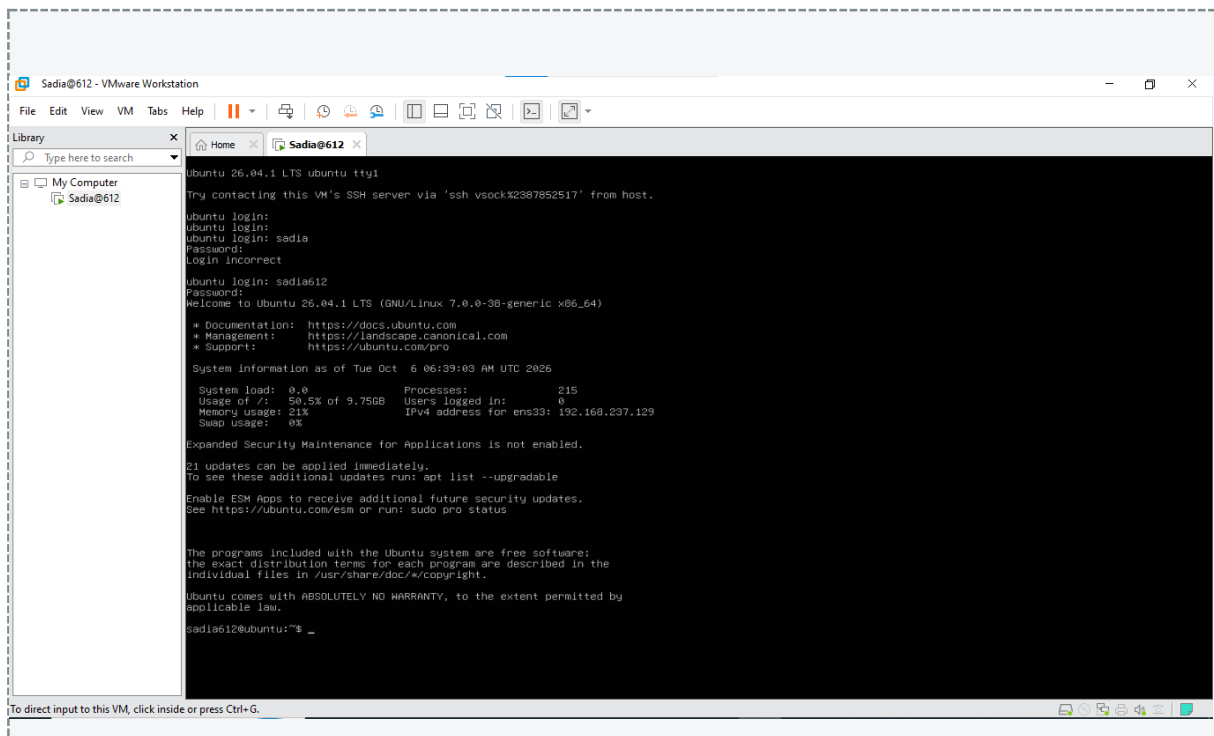


Figure: Ubuntu Server login screen after reboot.

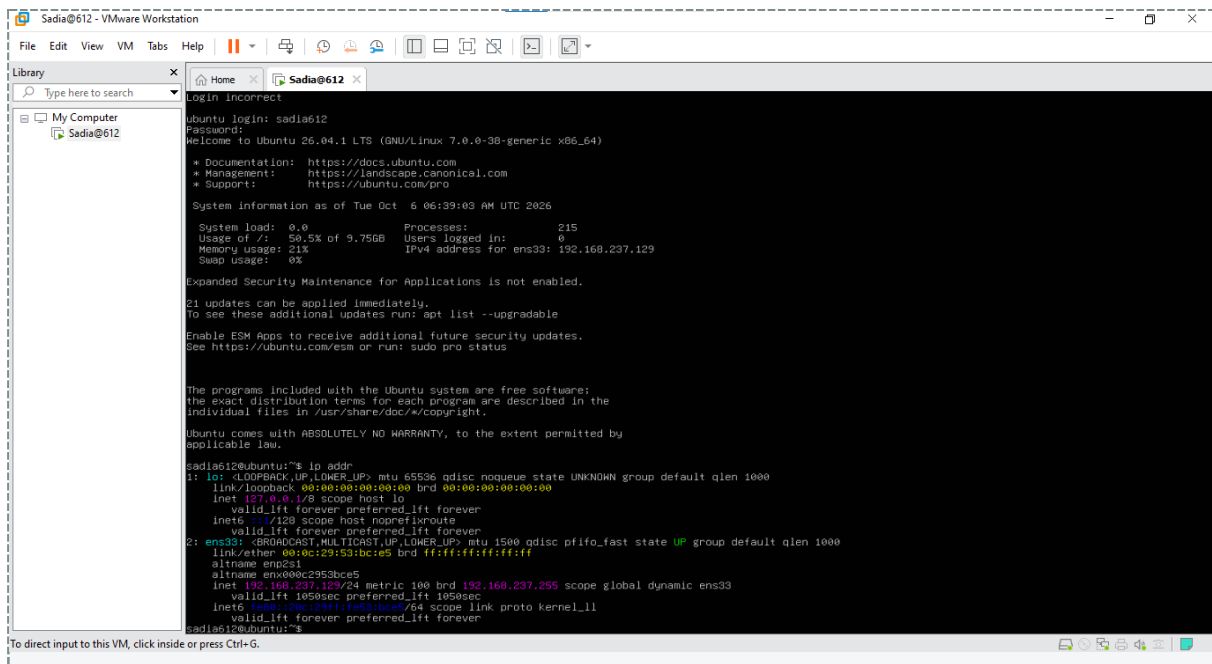


*Figure: Successful terminal login with my Ubuntu username and password.*



*Figure: Complete prompt in the format <github-username>@ubuntu:~\$.*

## Task 16: Find the Ubuntu Server IP Address



```
Sadia@612 - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Sadia@612
Login incorrect
ubuntu login: sadia612
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-38-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Tue Oct 6 06:39:03 AM UTC 2026

System load:  0.0          Processes:      215
Usage of /:   50.5% of 9.75GB   Users logged in:  0
Memory usage: 21%          IPv4 address for ens33: 192.168.237.129
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

21 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

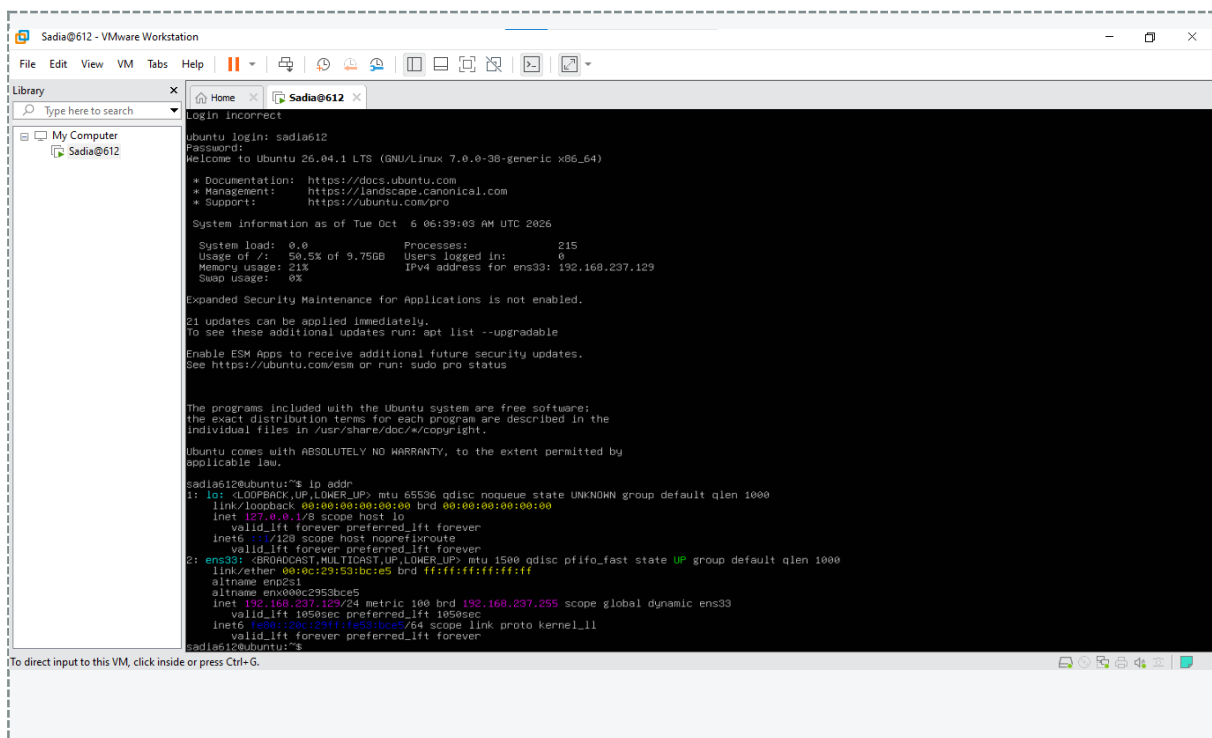
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

sadia612@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:53:bce5 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx00c2953bce5
    inet 192.168.237.129/24 metric 100 brd 192.168.237.255 scope global dynamic ens33
        valid_lft 180sec preferred_lft 180sec
    inet6 fe80::c29:53:bce5:64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
sadia612@ubuntu:~$
```

Figure: The ip addr command and its output inside Ubuntu Server.



```
Sadia@612 - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Sadia@612
Login incorrect
ubuntu login: sadia612
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-38-generic x86_64)

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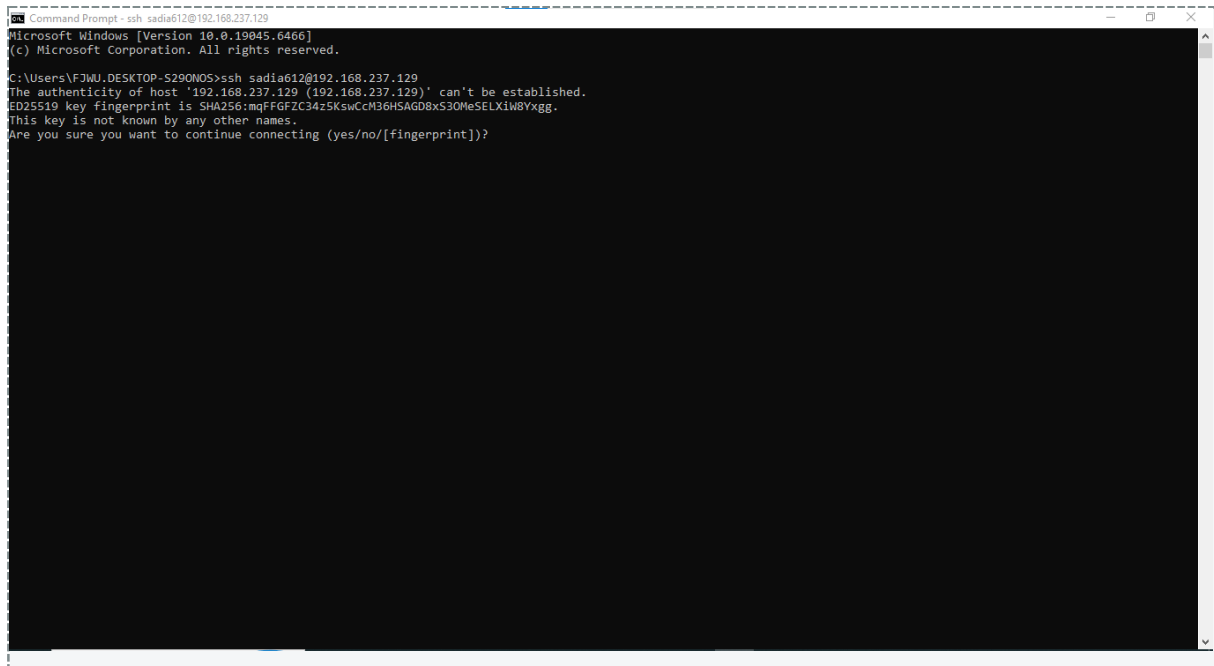
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the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

sadia612@ubuntu:~$ ip netns exec
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:53:bce5 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx00c2953bce5
    inet 192.168.237.129/24 metric 100 brd 192.168.237.255 scope global dynamic ens33
        valid_lft 180sec preferred_lft 180sec
    inet6 fe80::c29:53:bce5:64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
sadia612@ubuntu:~$
```

Figure: IPv4 address (after inet) of the active network interface, with the prompt visible.

## Task 17: Connect to Ubuntu Server from Windows



```
Command Prompt - ssh sadia612@192.168.237.129
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\FJMU-DESKTOP-S29ON05>ssh sadia612@192.168.237.129
The authenticity of host '192.168.237.129 (192.168.237.129)' can't be established.
ED25519 key fingerprint is SHA256:maFFGf7C34z5KswCcN3GH5AGD8xS3QMe5ELXiW8Yxgg.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])?
```

*Figure: SSH command run from Windows: `ssh <github-username>@<ubuntu-server-ip>`.*

```

sadia612@ubuntu: ~
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\FJWU.DESKTOP-S29ONOS>ssh sadia612@192.168.237.129
The authenticity of host '192.168.237.129 (192.168.237.129)' can't be established.
ED25519 key fingerprint is SHA256:mqFFGFZC34z5KswCcM36HSAGD8xS30MeSELXiW8Yxgg.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.237.129' (ED25519) to the list of known hosts.
sadia612@192.168.237.129's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-38-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Oct  6 06:49:15 AM UTC 2026

System load:  0.0           Processes:            219
Usage of /:   50.6% of 9.75GB Users logged in:        0
Memory usage: 21%          IPv4 address for ens33: 192.168.237.129
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

21 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

sadia612@ubuntu:~$

```

*Figure: Server fingerprint accepted by typing yes.*

```

sadia612@192.168.237.129's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-38-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
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To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

sadia612@ubuntu:~$

```

*Figure: Successful SSH login session (password not exposed).*

```

C:\Users\FJWU.DESKTOP-S29ONOS>ssh sadia612@192.168.237.129
The authenticity of host '192.168.237.129 (192.168.237.129)' can't be established.
ED25519 key fingerprint is SHA256:mqFFGFZC34z5KswCcM36HSAGD8xS30MeSELXiW8Yxgg.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.237.129' (ED25519) to the list of known hosts.
sadia612@192.168.237.129's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-38-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
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Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

sadia612@ubuntu:~$
sadia612@ubuntu:~$
sadia612@ubuntu:~$
```

*Figure: Remote prompt showing <github-username>@ubuntu.*